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Reflection on your internship to be included in your ePortfolio (500 words)

Where possible include picture of your internship illustrating the atmosphere of the internship or the specifics of your role.

What were your expectations about your internship before you joined?

I expected a challenge seeing as the area of work I entered relates to research into vibroacoustics However I also expected the workload to be reasonable with manageable goals seeing as this was an unpaid internship with my employers never having worked with interns before.

I was told to expect to learn how to develop and simulate phenomena in a virtual space using COMSOL Multiphysics and compare it to real life experimental results. In the process of doing this, I learnt the value of presentation skills, good data keeping habits, more "advanced" MATLAB scripting, and the challenges of matching virtual simulations to reality.

What was the reality? How was it different from your expectations?

Fortunately, every single learning goal and project "milestone" was reached, however due to the holiday break and supply-chain issues the experiment was delayed. This meant that collecting and learning about the experiment only had 2-3 weeks allocated to it, whereas the theoretical work and simulation jobs had several months. However the practical work that was conducted was very fruitful. The experimental setup involves measuring the vibrational response of the pipe at different locations to a hammer strike (see Figure 1).

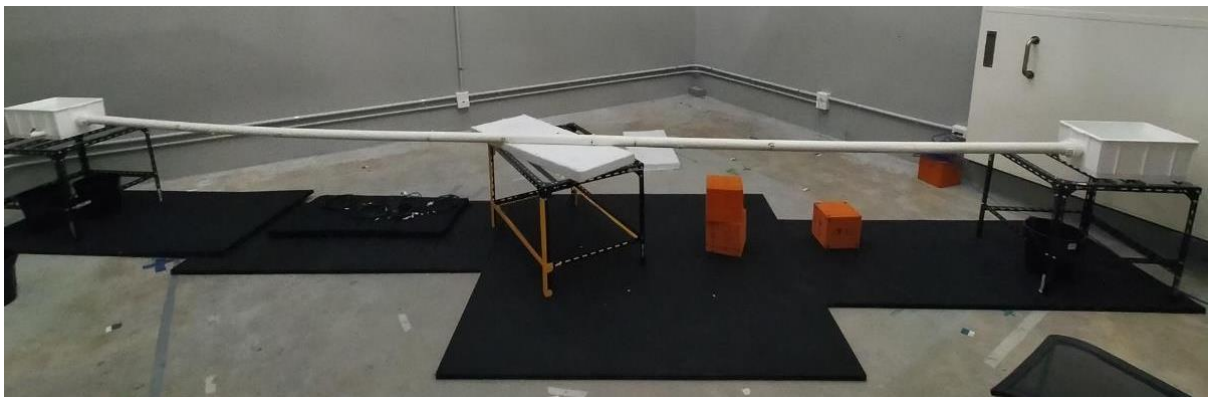

Figure 1. Experimental setup of the pipe acoustics study.

What lessons were the most important from your internship. Why were they important?

The most important engineering workplace skill learnt here was trying to model a digital twin of physical phenomena using high performance computing. As I worked with vibroacoustic researchers from the Centre for Audio, Acoustics, and Vibration, we

mainly focussed on vibration and frequency studies. I learnt how to implement Multiphysics of vibroacoustics, solidmechanics, and some thermodynamics during my internship to a relatively high standard (for an undergraduate student). To elaborate more on frequency studies (most relevant to the internship), I simulated the frequency response of a pipe against verified data (see image below –).

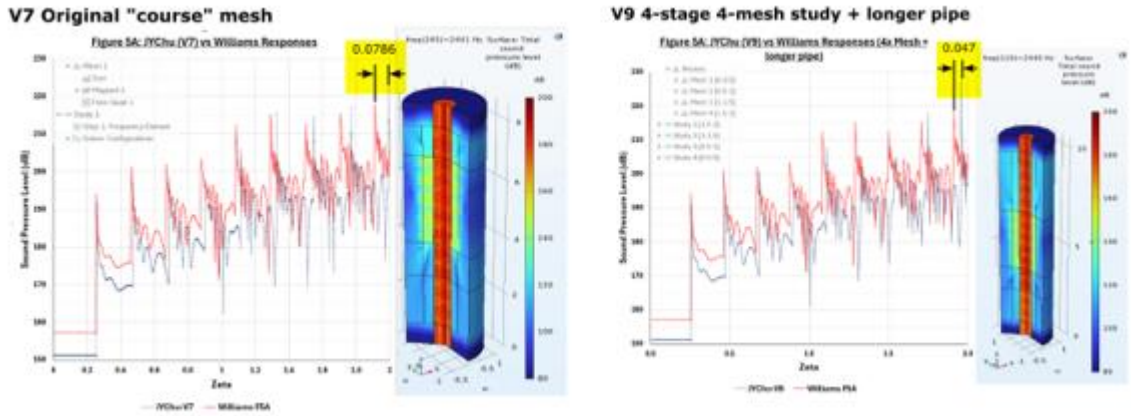

Figure 2. Shows a collage of frequency domain results collected from my simulation work using COMSOL. The plots showcase the sound pressure level response found at a particular point in the pipe. The black curve is the verified data, the red curve is my own data from simulations. The 3D model is a 3D representation of sound pressure level.

And later conducting experiments, I began learning how to measure time series data, process it, and convert to the frequency domain using fast Fourier transforms (see Figure 3).

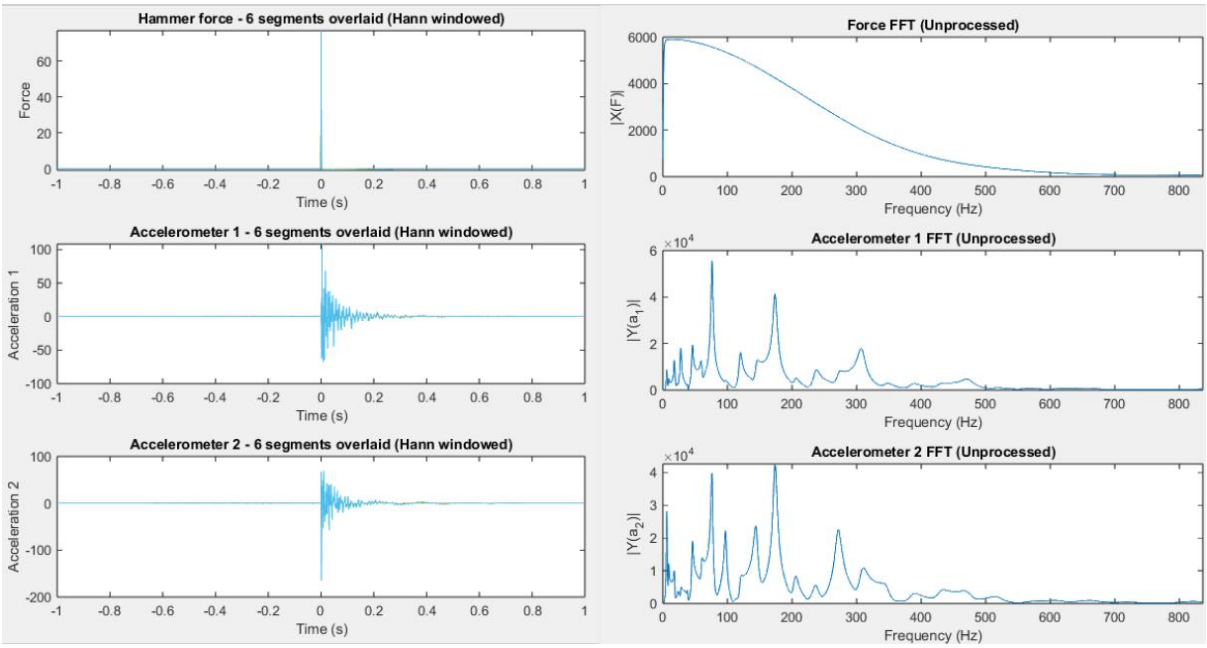

Figure 3. Data analysis using MATLAB, involving taking time series data and converting to the frequency domain using fast Fourier transforms.

Other lessons learnt from this internship are best summarised as dot points:

- **Importance of regular and clear communication** – our group met every 1-2 weeks, meaning that progress was expected each week with learning occurring every week. This meant learning occurred at a rapid pace.
- **Importance of file naming conventions** – files which weren't named in a "logical" way were difficult to decode at later stages.
- **MATLAB coding** – I learnt how to process signal data to produce a frequency response function in a meaningful way.
- **COMSOL modelling** – I learnt how to create a "digital twin" of a real life entity to simulate its Multiphysics behaviour (vibroacoustics, solid mechanics, and some thermodynamics).

After your workplace experience, what would you say your value proposition would be to an employer? How can you demonstrate this?

I provide a dedicated and detail-oriented engineering support with strong foundation in modelling, data analysis, and validation. Through my workplace experience I have demonstrated the ability to rapidly learn technical systems, problem solve efficiently, and provide accurate, well-documented results under time constraints. I add value through increasing technical clarity and ensuring outputs are reliable and fit for decision making.

How I demonstrate this:

- Delivering technically sound analyses (e.g., modelling, signal/data processing, verification against theory or standards).
- Producing clear documentation and structured reports that enable others to act confidently.
- Identifying errors, inefficiencies, or assumptions early and addressing them systematically.
- Managing tasks independently while maintaining quality and meeting deadlines.

How did your internship influence the type of role in which you are interested?

As of writing, I have agreed to undertake a part-time PhD with the same supervisors from the Centre for Audio Acoustics and Vibration. The internship greatly influenced my decision to pursue this opportunity. Additionally, and through my supervisor, I am exploring / discussing with a company to work part-time as a graduate Acoustic Engineer. I am grateful for the internship as without taking it, both options would not have opened up for me.